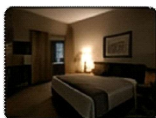




1. Appearance

modify the incoming frame before the policy

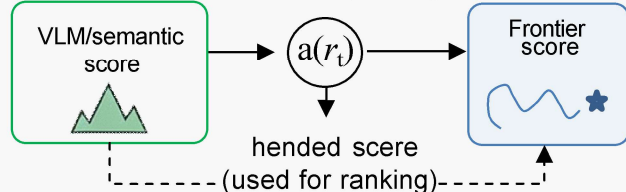
- **RES**: blanket restoration
- **OracleGote**: best-of-bound appearance upper bound



2. Where-to-go

reweight directional scoring under low reliability

- **M1g**: reliability-gated value-map reweighting



4. Forced commitment

force commitment when close-range evidence is available

- **CRV**: close-range recall verifier

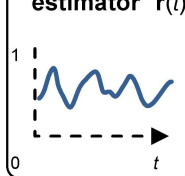


Degraded RGB observation

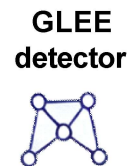


loss light/
motion blur/
fog/Geation noise

Blind reliability estimator $r(t)$



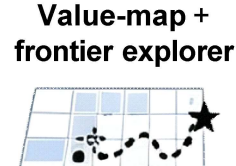
Frozen InstructNav backbone (clean-trained, frozen)



GLEE detector



Qwen2-VL-7B reasoning



Value-map + frontier explorer

Commitment bottleneck

single-frame, low-reliability commit

Outcome logs

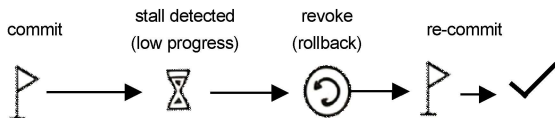
- reach
- commit
- stop/success

Diagnosis: degradation primarily breaks reach-to-success conversion rather than approach



3. Commit timing / revocation delay or retoohe unstable concitnents

- **REVOKE**: rerole staled commilment



5. Temporal / multi-view aggregation

accumulate evidence across frames or views

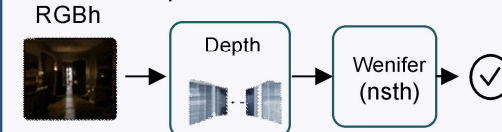
- **MUAP**: told-frame +re-observe
- **ROUTER**: pie-frerme dagodution router
- **FUSE**: multi-view identity fusion



6. Multimodal probe

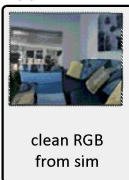
use an extra depth besd verifiar

- **DXCV**: dopt cos modal ernfer



Bench/Figure 4 : diagnostic instrument and stimuli

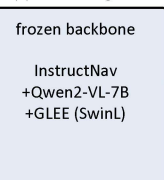
(a) Stimulus



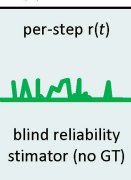
(b) Corruption



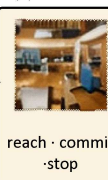
(c) Frozen agent



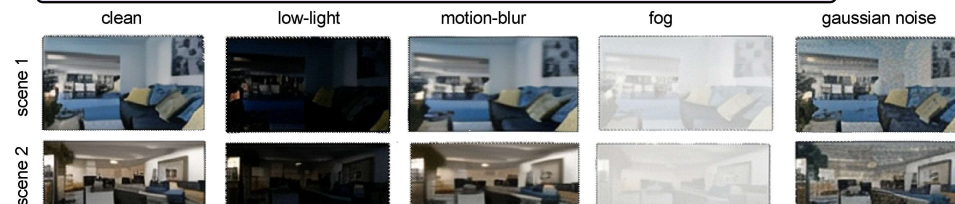
(d) Instrument



(e) Outcome



(f) Paired protocol: same seed and episode order across arms, N= 300 main split, McNemarexact p on discordant pairs, bootstrap 95% CI



Legend: where interventions act



All families are **test-time interventions** on a frozen clean-trained agent.